higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1: INTRODUCTION TO PLUMBING

1.1 In order to remove waste water. (2)

1.2
- Solar technology.
- Medical plumbing. (2)

1.3
- Colour
- Smell
- Flavour
- Solubility
- Boiling point
- Freezing point
- Expansion and contraction
- Density (Any 3 × 1) (3)

1.4 1.4.1 Water cycle is the process or cycle that water goes through when it turns from water into gas and then into water again.

1.4.2 Screening is the process used to treat raw water to separate dirt and impurities from the water.

(2 × 2) (4)

1.5
- Planning for the job.
- Briefing the plumber.
- Waiting in readiness.
- Overseeing the project. (4)

[15]

QUESTION 2: PLUMBING PRINCIPLES AND CODE OF PRACTICE

2.1 2.1.1 Surface water is the water that lies on top of the ground.

2.1.2 Ground water is water that lies among the layers of soil rock under the ground.

(2 × 2) (4)

2.2
- Well
- Borehole (2)

2.3
- The SANS – tested system.
- The joint accepted Scheme for water installation component.
- Agreement certificate. (3)

2.4 Pressure = Density x gravity x depth of the water (p=D x g x h). (2)

2.5
- It can be pulled – the pulling force of gravity.
- It can be pushed – with a pump. (2)

2.8 Flow rate is the amount of water that flows through a pipe in a certain amount of time. (2)
[15]

QUESTION 3: MATERIALS, TOOLS AND EQUIPMENT

3.1
3.1.1 F
3.1.2 D
3.1.3 A
3.1.3 C
3.1.5 B
(5 × 1) (5)

3.2
- Basin
- Bidets
- Baths
- Showers
- Sink
- Bath showers
- Shower tray

(Any 2 × 1) (2)

3.3 It is when water pipes vibrate, caused by badly fitted pipes and faulty washer. (2)

3.4
- Straight ladders
- Step ladders

(Any ONE) (1)
[10]

QUESTION 4: WATER PIPE ASSEMBLY

4.1
- Tape measure
- Hacksaw
- Reamer
- File stock
- Cutting oil
- Pipe wrench
- Pipe vice
- Marker/Pencil

(Any 5 × 1) (5)

4.2
- Capillary joints
- Compression joints

(2)

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4.3 1) First we use a spanner to undo the units that join the two elbows (A & C) to pipe work X & Y.
2) Remove the damaged pieces of pipe.
3) Cut off the compression ring so that you can remove nuts.
4) Cut a new piece of copper to fit the old pipe (B) fitted.
5) Ream the ends to get rid of burns.
6) Put the two nuts back on the pipe and fit new compression rings to both ends before you reassemble the elbow to the new pipe.
7) Refit the pipe to the pipe work that is still on the wall. (7)

4.4 Crimping is a special method used for joining multilayer pipes. (2)

4.5 4.5.1 True
4.5.2 False
4.5.3 True
4.5.4 True
4.5.5 False
(5 × 1) (5)

4.6
- Manually with your hands
- With an inside bending spring
- With an outside bending spring
- Using a bending tool (4)

[25]

QUESTION 5: CUTTING AND JOINING OF METALS

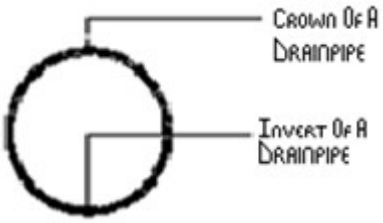
5.1 Tin snip (1)

5.2 Flashing is a strip of material that will make a join between the roof and brickwork water proof. (2)

5.3. 5.3.1 Skirting
5.3.2 Soakers
5.3.3 Zinc
5.3.4 Mitre
5.3.5 10 mm
(5 × 1) (5)

5.4
- Brazing
- Welding (2)

[10]



3 L2

QUESTION 6: SET OUT BELOW GROUND DRAINAGE SYSTEMS

6.1
- There should be enough pipes to lead the wastewater to the system.
- The pipes must be as straight as possible between the places where they will be reached or accessed.
- Pipes must be properly supported so that they don't sag and cause dips in the middle or pull out of their joins.
- Pipes must be laid so that the water will flow down and through them easily.
- The joins and connections as well as the inspection must be water tight.
- Drains should never run under a building. (6)

6.2 A gradient is a slope or the degree of a slope. (1)

6.3

✓

✓ ✓ (3)

6.4
6.4.1 North Point
6.4.2 Centre line
6.4.3 Datum level
6.4.4 Invert level
6.4.5 Diameter

(5 × 1) (5)

6.5 Manholes are chambers that are at least 750 mm deep and large enough for a person to climb into. (2)

6.6
- Concrete bedding
- Clean and sieved building sand
- Selected granular material
- Non organic material (Any 2 × 1) (2)

6.7 Temporary timbering is used to support the sides of trenches to stop them from covers collapsing while people are working in the trenches. (2)

6.6
- Water level
- Straight edge with spirit level
- Gradient board for short drains
- Dumpy level
- Boning rod
- Sight rail
- Theodolite (Any 4 × 1) (4)

[25]

TOTAL: 100

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